

Experiment-6

Vacuum Cleaner Problem

Aim:

To develop a Python program to solve the Vacuum Cleaner Problem using a Simple Reflex Agent in Artificial Intelligence.

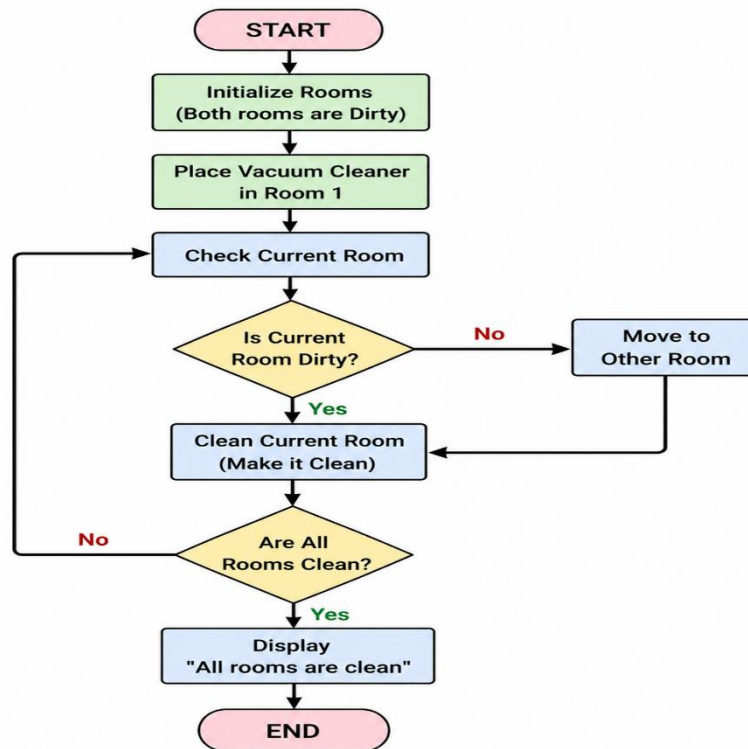
Objective:

- To simulate the working of a vacuum cleaner agent.
- To clean all dirty rooms using simple decision rules.
- To understand the concept of intelligent agents in Artificial Intelligence.

Algorithm:

1. Start the program.
2. Initialize the rooms as dirty.
3. Place the vacuum cleaner in the first room.
4. Check the current room.
5. If the room is dirty, clean it.
6. Otherwise, move to the other room.
7. Repeat until all rooms are clean.
8. Display that all rooms are clean.
9. Stop the program.

Flowchart:



Code:

```
rooms = ['Dirty', 'Dirty']
```

```
location = 0
```

```
while 'Dirty' in rooms:
```

```
    if rooms[location] == 'Dirty':
```

```
        print("Cleaning Room", location + 1)
```

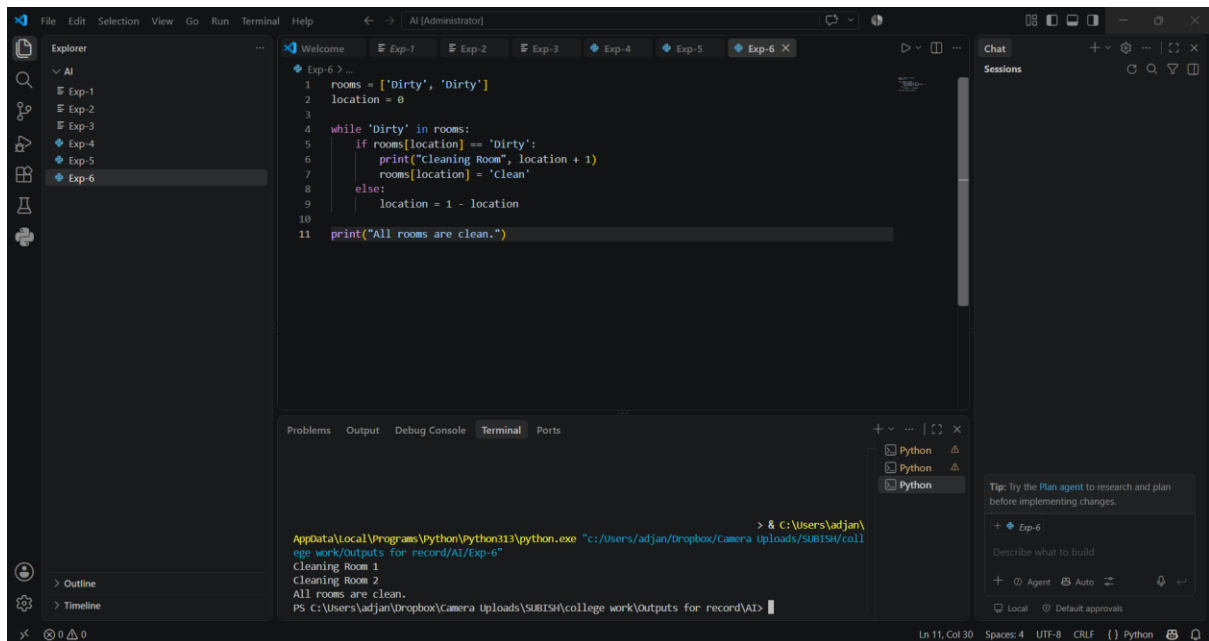
```
        rooms[location] = 'Clean'
```

```
    else:
```

```
        location = 1 - location
```

```
print("All rooms are clean.")
```

Output:



The screenshot shows a Visual Studio Code editor with a Python script for the Vacuum Cleaner Problem. The script is named 'Exp-6' and is located in the 'AI' folder. The script defines a list of rooms, a current location, and a while loop that continues until all rooms are clean. The terminal output shows the execution of the script, which prints 'Cleaning Room 1', 'Cleaning Room 2', and 'All rooms are clean.'

```
1 rooms = ['Dirty', 'Dirty']
2 location = 0
3
4 while 'Dirty' in rooms:
5     if rooms[location] == 'Dirty':
6         print("Cleaning Room", location + 1)
7         rooms[location] = 'Clean'
8     else:
9         location = 1 - location
10
11 print("All rooms are clean.")
```

The terminal output shows the execution of the script, which prints 'Cleaning Room 1', 'Cleaning Room 2', and 'All rooms are clean.'

```
> & (C:\Users\adjan\
AppData\Local\Programs\Python\Python313\python.exe "c:\Users\adjan\Dropbox\Camera uploads\SUBISH\college work\Outputs for record\AI\Exp-6"
Cleaning Room 1
Cleaning Room 2
All rooms are clean.
PS C:\Users\adjan\Dropbox\Camera uploads\SUBISH\college work\Outputs for record\AI>
```

Result:

The Vacuum Cleaner Problem was successfully solved using a Simple Reflex Agent, and all dirty rooms were cleaned.

Conclusion:

The Vacuum Cleaner Agent efficiently cleaned all rooms by following simple condition-based rules, demonstrating the basic concept of intelligent agents in Artificial Intelligence.